

A Project Report on
Zoo Management System

Submitted to
DR. BABASAHEB AMBEDKAR TECHNOLOGICAL
UNIVERSITY, LONERE

in partial fulfillment of the requirement for the degree of
BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & ENGINEERING
By

Pranav Lambe
Omsai Kadam
Shubham Kure

Under the Guidance
of
Ms. Shelke M. G.

(Department of Computer Science and Engineering)



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MAHATMA GANDHI MISSION'S COLLEGE OF ENGINEERING
NANDED (M.S.)
Academic Year 2020-21

Certificate



This is to certify that the project entitled

“Zoo Management System ”

being submitted by Mr. Pranav Lambe, Mr.Omsai Kadam Mr.Shubham Kure to the Dr. Babasaheb Ambedkar Technological University, Lonere, for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is a record of bonafide work carried out by him/her under my supervision and guidance. The matter contained in this report has not been submitted to any other university or institute for the award of any degree.

Ms. Shelke M. G.
Project Guide

Dr. A. M. Rajurkar

H.O.D

Computer Science & Engineering

Dr. Ms. G. S. Lathkar

Director

MGM's College of Engg., Nanded

TABLE OF CONTENTS

Acknowledgement	I
Abstract	II
Table of Contents	III
List of Figures	V
List of Tables	VI
Chapter 1. INTRODUCTION	1
1.1 Zoo Introduction	1
1.2 Need for a Zoo Management System	3
1.3 Problem Statement	3
Chapter 2. LITERATURE REVIEW & THEORETICAL BACKGROUND	19
2.1 Management Information System	19
2.2 Database Management System	19
2.3 Software Development Life Cycle	20
Chapter 3. SYSTEM ANALYSIS AND DESIGN	33
3.1 Architecture client-server	33
3.2 Context Diagram	34
3.3 Database Architecture	44
Conclusion	48
References	50

ABSTRACT

The Zoo Management System is a software solution designed to simplify and automate the operations of a zoo. It manages essential information such as animal records, habitats, feeding schedules, medical treatments, staff details, and visitor management. By replacing traditional manual processes with a centralized digital platform, the system improves data accuracy, enhances coordination, and reduces administrative workload.

The system allows administrators and zookeepers to efficiently track animal health, dietary needs, and daily activities, while veterinarians can monitor medical histories and schedule timely treatments. Staff duties, ticketing operations, and maintenance activities are also managed systematically to ensure smooth workflow. Through an integrated database and automated modules, the Zoo Management System improves decision-making, enhances animal care, and provides a better experience for both staff and visitors. This technology-driven approach makes zoo administration more efficient, transparent, and reliable.

ACKNOWLEDGMENT

We are greatly indebted to our Project guide, **Ms. Shelke M. G.** for her able guidance, and we would like to thank her for her help, suggestions, and numerous helpful discussions.

We gladly take this opportunity to thank **Dr. A.M. Rajurkar** (Head of Computer Science and Engineering, MGM's College of Engineering, Nanded).

We are heartily thankful to **Dr. G. S. Lathkar** (Director, MGM's College of Engineering, Nanded) for providing facilities during the progress of the Project and for her kind guidance and inspiration.

Last but not least, we are also thankful to all those who helped directly or indirectly in the complete and successful development of this Project.

With Deep Reverence,

Pranav Shrikant Lambe [201]
Shubham Padmakar Kure [220]
Omsai Gangadhar Kadam[225]

INTRODUCTION

1.1 Zoo Introduction

Zoos represent an indispensable component of global wildlife conservation and environmental sustainability. They act as protective habitats where various species, especially endangered ones, are carefully preserved, rehabilitated, and studied. Beyond conservation, modern zoos serve as research laboratories for biological studies, centers of public awareness, and institutions promoting ethical treatment of animals.

With increasing global concern regarding species extinction, climate change, and habitat destruction, the role of zoos has expanded from mere recreational spaces to institutions responsible for protecting biodiversity and educating society about ecological balance.

As zoo operations expand, the administrative responsibilities associated with managing animals, staff, visitors, veterinary care, feeding schedules, and finances become progressively complex. Manual, paper-based systems are no longer capable of handling the volume and complexity of data required for day-to-day operations. Errors, delays, information gaps, and inefficiency can compromise both animal welfare and organizational performance. Therefore, there is a growing need for digital transformation in zoo administration.

The Zoo Management System (ZMS) emerges as a modern software-based solution designed to automate and integrate all essential functions of a zoo. It streamlines the process of recording animal data, tracking medical history, scheduling feed intervals, organizing staff duties, administering visitor ticketing, and managing financial records. The use of such a system ensures greater accuracy, reduces paperwork, eliminates duplication of information, and improves decision-making by providing real-time and historical data accessibility. Ultimately, a well-designed ZMS enhances operational efficiency, improves transparency, and supports the strategic goals of conservation and education.

1.2 Need for a Zoo Management System

The traditional approach to zoo management depended heavily on handwritten registers, informal notes, excel files, and manually updated record books. While such methods may have been adequate in earlier times, they are now insufficient due to the increased scale of operations and the demand for precision. When data is dispersed across multiple locations and formats, retrieving information becomes time-consuming and prone to error. Inaccurate or misplaced records can result in mistimed feeding schedules, neglected medical treatments, or poor visitor service, ultimately affecting both animal welfare and organizational credibility.

Furthermore, modern zoos require timely reports for internal evaluation, government compliance, conservation programs, and financial auditing. Manual systems struggle to generate analytical insights or produce structured reports on animal health, breeding patterns, staff performance, or visitor statistics. Therefore, automation is essential to replace fragmented and outdated systems, enabling efficient workflow management and informed administrative decisions.



1.3 Problem Statement

Zoo operations involve numerous activities that must function in a coordinated and structured manner. Managing diverse processes such as animal identification, species classification, feeding schedules, medical treatments, habitat allocation, staff coordination, ticketing services, income tracking, and visitor flow presents significant challenges when handled manually. Paper-based records are vulnerable to damage, loss, and human error. Searching for past information, especially medical or breeding history, can take substantial time, which may lead to delays during emergencies.

Lack of integration between departments such as administration, veterinary services, ticketing, and caretaking results in fragmented information flow and operational inefficiency. Since critical decisions rely on accurate data, any inconsistency can negatively impact animal safety and organizational functioning. In this context, there is a clear requirement for a computerized Zoo Management System capable of centralizing all records, ensuring fast retrieval of information, and maintaining accurate documentation in a structured and secure manner.

1.4 Objectives of the Project

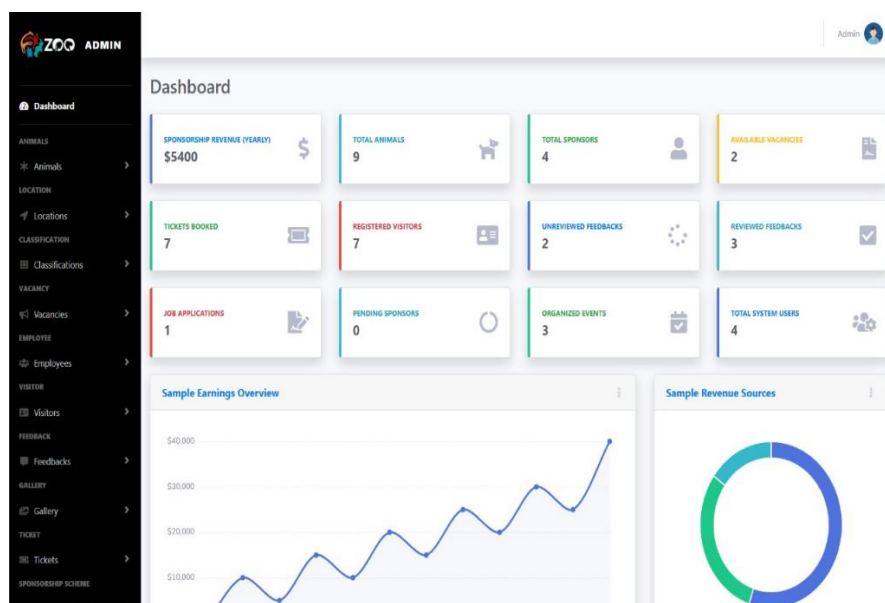
The primary objective of developing the Zoo Management System is to facilitate seamless and automated handling of day-to-day zoo operations through a centralized digital platform. It aims to reduce dependency on manual documentation and enhance the efficiency and reliability of operations. A key goal of the system is to support animal welfare by ensuring accurate tracking of feeding and medical schedules, enabling timely care interventions. The system also intends to improve administrative processes by providing structured access to data for different users such as administrators, veterinarians, caretakers, and ticketing personnel.

Another important objective is to promote resource optimization by reducing redundancy, minimizing errors, and supporting informed managerial decisions through analytical reports. The system also enhances visitor interactions by improving ticket processing and record management. Ultimately, the Zoo Management System aims to support sustainable

and well-regulated zoo operations in alignment with modern technological and conservation standards.

1.5 Scope of the Project

The proposed Zoo Management System covers the digital management of animal data, feeding and medical scheduling, staff coordination, and visitor ticketing. It provides functionalities for



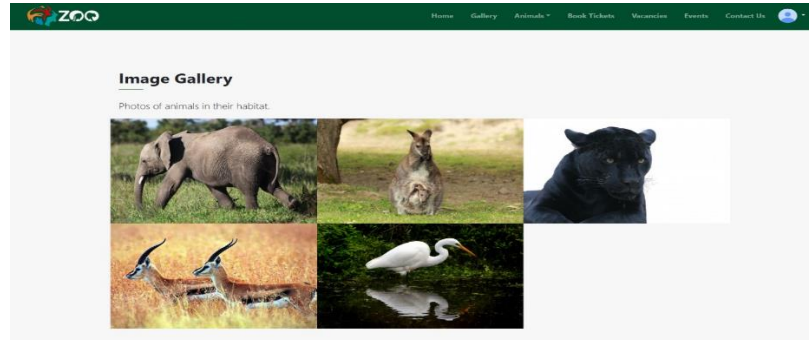
adding, updating, monitoring, and retrieving records related to species information, enclosure assignments, healthcare treatments, and nutrition management. It also facilitates the administration of employee roles, duty schedules, and work allocation, while providing systematic recording of visitor details and financial transactions. Additionally, the system offers automated report generation for operational, administrative, and financial analysis to support long-term planning and development.

The project focuses primarily on software-based solutions and does not extend to hardware-based automation such as sensor-controlled feeding mechanisms, robotic surveillance, or IoT-driven monitoring tools. It provides an efficient management framework rather than a physical automation system.

1.6 Importance of Zoo Management Systems

Implementing a Zoo Management System significantly contributes to improving overall zoo performance. It supports conservation initiatives by enabling accurate tracking of species records, breeding programs, and veterinary treatments. The system ensures improved coordination among departments, helping to maintain high animal welfare standards and enhancing response preparedness during health emergencies. Digital documentation promotes transparency and provides reliable evidence for audits, regulatory compliance, and scientific research collaborations.

For visitors, the system enhances the experience by reducing waiting time and increasing accessibility to services. For administrators, it provides analytical insights that guide decision-making and resource planning. In the broader context, ZMS assists zoos in aligning with global technological trends, enabling them to meet international standards of wildlife care and data management.



1.7 Organization of the Report

This report consists of five chapters. Chapter 1 introduces the study, defines the need for the project, and outlines the problem scope and objectives. Chapter 2 presents the theoretical foundation and discusses previous research and existing systems. Chapter 3 explains the system requirements, current system analysis, and proposed system design including diagrams and database structure. Chapter 4 describes the implementation

process, development tools, and integration of various modules. Finally, Chapter 5 covers the testing methods, results, discussion, conclusions, and future enhancement possibilities.

CHAPTER 2

LITERATURE VIEW & THEORETICAL BACKGROUND

The literature review is a critical component of any academic research project, providing a foundation upon which the proposed system is conceptualized and developed. It offers insights into previously conducted studies and technological advancements relevant to the Zoo Management System. By understanding the principles and models associated with information systems, database management, software development methodologies, and wildlife conservation technologies, it becomes possible to justify the need for systemic automation in zoo administration. The review explores scholarly theories and practical approaches that contribute to designing a robust digital platform for managing zoo operations efficiently.

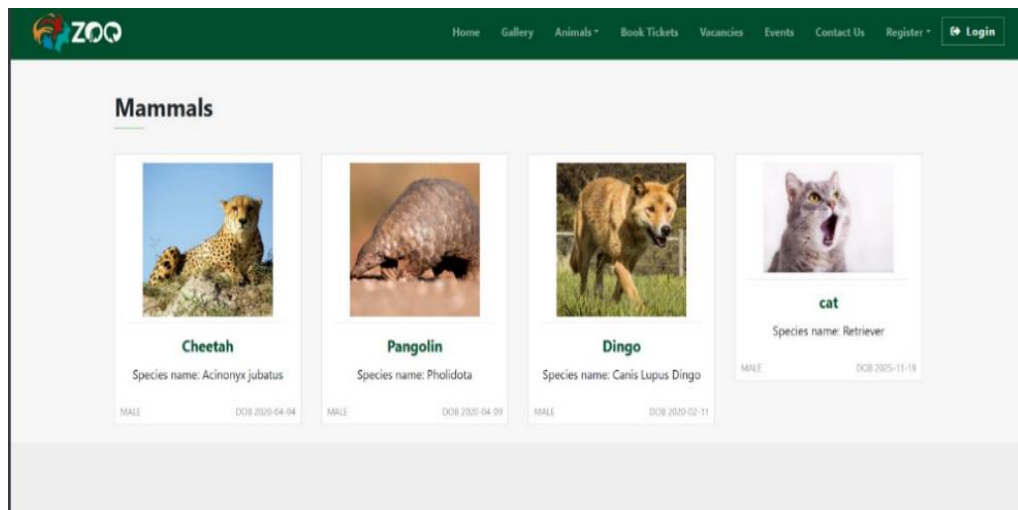
2.1 Management Information System

Management Information Systems represent a well-established research field focusing on the application of computer-based tools to support managerial functions, decision-making, and organizational control. MIS integrates computer science, management theory, and decision sciences to improve efficiency and enhance data-driven decision-making. According to MIS theory, information is considered one of the most valuable organizational resources, comparable to materials, finance, and manpower. Effective management depends on structured data collection, processing, storage, and dissemination.

In the context of zoos, MIS assists in maintaining complex organizational processes involving biological research, veterinary services, staff coordination, and visitor management. Since zoo operations involve dynamic and sensitive variables such as animal health indicators, feeding intervals, and emergency treatments, MIS ensures accuracy and reliability. It enables rapid access to information that managers require for planning conservation strategies, assigning resources, and complying with legal requirements related to wildlife protection. A Zoo Management System therefore functions as an applied MIS tailored specifically for zoological environments, integrating diverse operational units into a unified structure supported by timely and accurate information flow.

2.3 Database Management System

The Database Management System is fundamental to large-scale digital information storage and retrieval. DBMS supports the systematic organization of structured data using relational models, tables, constraints, and queries. In zoo administration, thousands of data entries must be preserved, including animal identification codes, genetic background, disease histories, breeding records, medical evaluations, nutritional plans, and ticketing transactions. Traditional manual systems lead to fragmented and inconsistent records, making DBMS essential for integrity and reliability.



DBMS theory emphasizes principles such as data independence, data security, redundancy elimination, transaction management, concurrency control, and backup and recovery. These principles ensure that data remains consistent even when accessed simultaneously by different stakeholders such as veterinarians, caretakers, researchers, administrators, and ticketing staff. Through normalization, DBMS prevents duplication of information and ensures efficient structuring of data relationships. The ability to generate complex queries and real-time reports assists in scientific studies, monitoring endangered species, auditing operations, and evaluating visitor patterns. Without DBMS integration, zoo operations face higher risks of operational failure, data loss, and poor analytical output.

2.4 Software Development Life Cycle

The Software Development Life Cycle is a conceptual model that guides systematic software creation and ensures that the final product meets both functional and non-functional expectations. SDLC includes sequential phases of requirement analysis, system design, implementation, testing, deployment, and maintenance. The Waterfall model, one of the earliest and most structured SDLC frameworks, is especially suitable for the development of the Zoo Management System because its requirements are clearly defined, predictable, and stable.

The Waterfall approach ensures high-quality documentation, reducing ambiguity and improving communication between developers and stakeholders. Each stage is thoroughly validated before progressing to the next, reducing the chances of major structural revisions later. While modern software industries often adopt Agile or Spiral models for rapidly changing projects, a zoo system is typically driven by procedural consistency rather than rapidly evolving requirements. Thus, the Waterfall model aligns effectively with the linear and standardized nature of zoo processes such as animal record keeping, ticketing, and staff management.

2.5 Information Flow in Zoo Management

The flow of information within a zoo determines the effectiveness of communication and operational performance. Zoos involve various interdependent departments, and any delay or inaccuracy in communication can negatively impact animal welfare. For example, medical staff must receive immediate alerts about scheduled vaccinations or disease symptoms, while caretakers must execute feeding instructions at precise intervals based on nutritional standards. Ticketing data must be quickly transferred to finance departments for revenue tracking and planning.

In a manual environment, information is often dispersed across disparate systems, leading to delays, inconsistencies, and loss of critical details. Digital automation, on the other hand, enables centralized storage and circulation of data, ensuring that updates are transmitted instantly and uniformly. Proper information flow reduces human error and enhances transparency, contributing to professional and ethical management practices. It also

provides valuable insights for regulatory agencies, zoological associations, and scientific collaborations.

2.6 Role of ICT in Wildlife Conservation

Information and Communication Technology has transformed the field of environmental science and wildlife conservation. ICT tools facilitate long-term scientific research, advanced species monitoring, genetic database management, remote sensing technologies, GPS tracking, camera surveillance systems, and predictive analytics. Zoos worldwide rely on digital systems to participate in global conservation networks such as Species Survival Plans (SSP), international breeding databases, and collaborative research programs.

Advances in technology have enabled veterinarians to monitor health indicators using AI-based diagnostics and machine learning models. Data archiving plays a vital role in preventing species extinction by identifying breeding compatibility, monitoring hereditary diseases, and analyzing behavioral changes. ICT's contribution goes beyond administrative efficiency—its broader impact lies in preserving biodiversity and reinforcing global conservation ethics. Hence, the Zoo Management System becomes a gateway through which local zoos contribute to global scientific knowledge and wildlife sustainability.

2.7 Existing Zoo Management Systems

A review of existing digital zoo management solutions demonstrates the importance of centralized data systems. Highly recognized platforms such as ZIMS (Zoological Information Management System) developed by Species360 are utilized in more than 1,000 zoological institutions worldwide. ZIMS provides standardized global databases that supply detailed species records to researchers, enabling collaborative conservation strategies. Similarly, ARKS (Animal Record Keeping System) is widely used for maintaining historical records for both captive and endangered wildlife species.

Despite the availability of these advanced systems, many small or government-run zoos do not adopt them due to cost, licensing restrictions, limited technical expertise, or infrastructural constraints. This gap indicates a strong need for practical, affordable, and scalable systems like the proposed Zoo Management System. It highlights the potential of

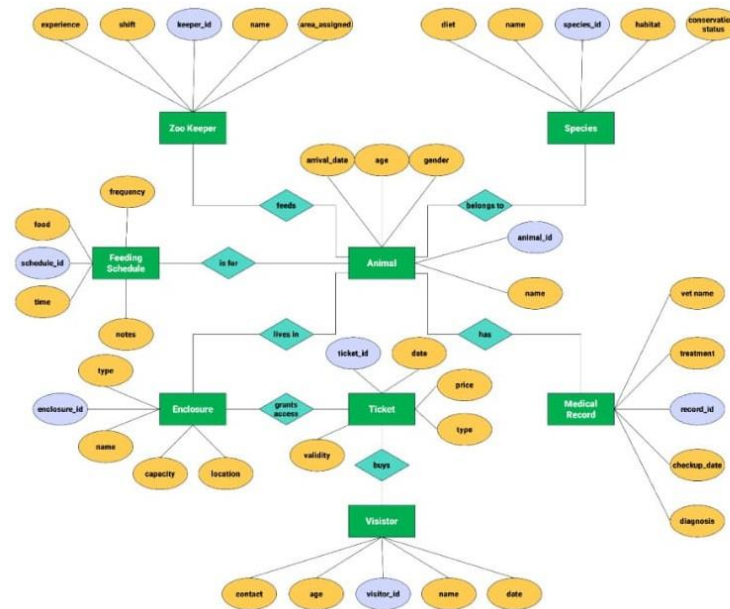
developing customized solutions suited to local operational requirements, rather than relying solely on expensive global frameworks.

2.8 Summary

The literature review confirms that the adoption of digital management systems is critical for modern zoological operations. MIS provides the conceptual foundation for organizing and controlling organizational processes, while DBMS ensures secure and structured data storage. SDLC frameworks guide software development in a systematic and organized manner, ensuring that user needs are met effectively. ICT supports long-term conservation goals and international research collaboration. The study of global zoo management systems reinforces the importance of digital transformation and validates the need for implementing a comprehensive Zoo Management System. Together, these theories and existing technologies form the intellectual and practical basis upon which the proposed project is built.

SYSTEM ANALYSIS AND DESIGN

System Analysis and Design is an essential phase in the development of the Zoo Management System because it helps in understanding the existing working conditions, identifying the problems within the current manual system, and proposing an effective computerized solution. The analysis stage focuses on studying how zoo records are currently maintained, how data is collected and processed, and what difficulties users face while accessing information. Traditionally, many zoos maintain records manually in registers or simple spreadsheets, which makes updating, retrieving, and managing information extremely time-consuming and prone to human errors. This manual approach makes it difficult to track animals' medical histories, feeding schedules, vaccination records, staffing details, and visitor ticketing. As a result, data duplication, loss of crucial information, low data security, and slow report generation are common challenges in the



To overcome these problems, the proposed Zoo Management System introduces a fully computerized and centralized database solution that stores, processes, and retrieves

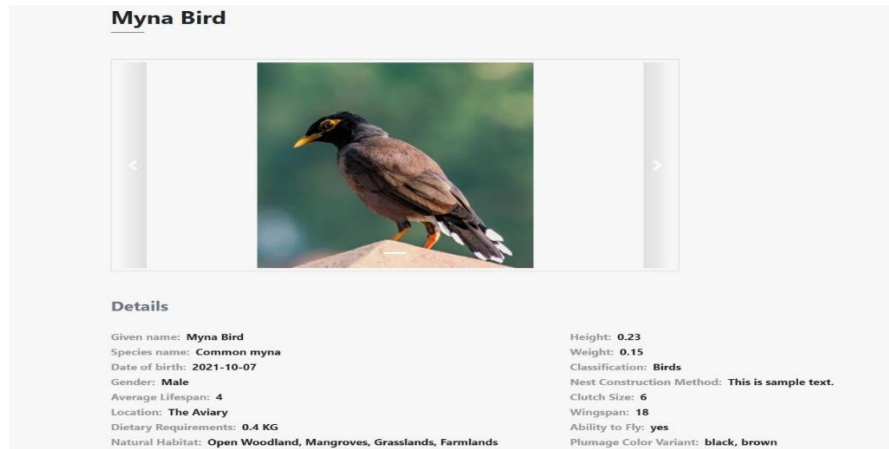
information efficiently. The system helps in improving accuracy, reducing paperwork, enhancing security, and enabling real-time monitoring of animals and zoo operations. It supports important features such as automated ticket booking, report generation, staff and animal management, and role-based secure access for admin, staff, and visitors. Through this proposed system, users can easily maintain and update records without errors and access data quickly whenever needed.

3.1 Architecture client-server

The design phase of the system converts collected requirements into a structured blueprint that guides the development process. System design defines the architecture, modules, interface layout, and database structure of the application. The logical design includes models such as Data Flow Diagrams, ER diagrams, and Use Case diagrams to illustrate the system workflow and interaction between different entities. The physical design, on the other hand, focuses on the actual implementation details such as hardware needs, software environment, database tables, and user interface components. The Zoo Management System is developed using a client-server architecture where the user interacts with the front-end interface, the application server processes requests, and the database stores all information securely.

3.2 Context Diagram

The Data Flow Diagram (DFD) Level 0, also known as the Context Diagram, represents the overall structure of the Zoo Management System and illustrates how external entities interact with the central system. In this diagram, the Zoo Management System is depicted as a single main process, shown in the center, which receives data from and sends data to multiple external users.



Admin:- provides system input such as managing user accounts, controlling access permissions, and updating critical information related to animals, staff, or financial records. The system processes these inputs and stores them in the internal database.

Staff:- interacts with the system to perform daily operational tasks such as updating feeding schedules, submitting animal status reports, and handling ticketing activities. The system then processes this information and offers relevant updates and reports as output. Visitor interacts with the system mainly for ticket booking and entry registration. Visitors receive services such as ticket confirmation, pricing details, and booking status generated by the system.

Veterinarian:- provides medical updates, vaccination records, and treatment history for animals. In return, the system provides access to animal health reports, medical history, and alerts for scheduled treatments.

Thus, DFD Level 0 provides a high-level overview of the information flow, showing how input from Admin, Staff, Visitors, and Veterinarians is processed and how useful output is delivered back to the respective stakeholders. At this level, no internal processes, data stores, or submodules are shown; instead, the focus remains on external interactions and overall system boundaries.

3.3 Database Architecture

Database Architecture is the logical framework that describes how a database system is structured and how various components communicate with each other. It acts as a blueprint

for designing and implementing a DBMS. A well-designed architecture ensures smooth data management, efficient storage, high performance, and secure access. It separates the physical storage of data from the user view, which means users do not need to know where or how data is physically stored. Instead, they interact with the database using applications and interfaces provided by the architecture.

The architecture is mainly categorized into Single-Tier, Two-Tier, and Three-Tier models. Each layer has a specific purpose such as user interaction, processing of business logic, and data storage. This layered approach improves modularity and flexibility, making it easier to scale and maintain systems as organizational requirements grow. Modern DBMS such as Oracle, MySQL, SQL Server, and PostgreSQL are based on these architectural models to support efficient data handling.

In Single-Tier Architecture, also known as standalone architecture, the user interacts directly with the database without any network or server involvement. All operations occur on the same machine. It is mainly used in educational learning, small offices, and personal database projects where limited storage and fewer users exist. Although simple and less expensive, it lacks security, backup mechanisms, and multi-user access.

Two-Tier Architecture, or client-server architecture, divides work between client systems and a database server. The client sends requests using SQL queries, and the server processes and returns the results. This architecture provides improved performance, better security, and supports multi-user environments. Examples include college management systems, small company applications, and local banking systems.

Three-Tier Architecture is the most advanced and widely used model in modern enterprise systems. It consists of three layers: Presentation Layer (client interface), Application Layer (business logic), and Database Layer (data storage). The application server processes complex operations, reduces the load on the database, and improves system security. It is suitable for large, distributed applications like online shopping, social media, healthcare information systems, and cloud-based services.

CONCLUSION

The engineering project on RFID-Based Automated Attendance System has demonstrated a modern, efficient, and reliable method of maintaining attendance records in educational institutions and workplaces. The system reduces the dependency on manual processes and eliminates common errors such as proxy attendance, time manipulation, and record mismanagement. By using RFID cards and readers, the process becomes fast, contactless, and user-friendly, ensuring smooth automation. This project successfully integrates hardware components such as the RFID reader, microcontroller, and buzzer with a software interface for secure data storage. The recorded attendance can be monitored in real-time and stored safely in a database for future reference. The system proves to be cost-effective and scalable, meaning it can be expanded to handle large numbers of users with minimal modifications.

REFERENCES

- K. Finkenzeller, *RFID Handbook: Fundamentals and Applications in Contactless Smart Cards, Radio Frequency Identification and Near-Field Communication*, 3rd ed. Hoboken, NJ, USA: Wiley, 2010. ISBN: 978-0-470-69506-7.
- M. A. Mazidi, S. Naimi, and S. Naimi, *AVR Microcontroller and Embedded Systems: Using Assembly and C*, Upper Saddle River, NJ, USA: Pearson Education, 2011. ISBN: 978-0-13-800331-9.
- R. S. Gaonkar, *Microprocessor Architecture, Programming and Applications with the 8085*, 6th ed. Delhi, India: Penram International Publishing, 2013. ISBN: 978-81-904526-5-2.
- R. P. Jain, *Modern Digital Electronics*, 4th ed. New Delhi, India: McGraw-Hill, 2010. ISBN: 978-0-07-066911-6.